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Research Statement

My research interests: Empirical Industrial Organization, Applied Microeconometrics.

My research uses microdata and quantitative methods to study firm productivity and firm and industry dynamics. I also work on structural econometric modeling, with a focus on the identification of production functions. I examine how technology adoption changes production, input choices, and market power. A central challenge is that firms choose inputs using information about productivity that researchers do not observe. My work addresses this endogeneity to distinguish technological improvements from other sources of firm performance.

My agenda connects empirical studies of manufacturing and banking with econometric research on causal mechanisms and heterogeneous treatment effects. It includes a paper published in *Economics Letters* and a paper published in *Economic Modelling*. Across these projects, economic models guide both the measurement of productivity and the interpretation of empirical findings.

Job Market Paper: Digitalization, Productivity, and Market Power

My job market paper, “**Digitalization, Factor-Biased Productivity, and Market Power**” (with Subal C. Kumbhakar), asks what digitalization changes inside a firm. Digital technologies may make labor more effective, improve productivity more broadly, or alter a firm’s ability to charge prices above marginal cost. These possibilities have different implications for workers, firms, and competition, yet a single productivity measure can obscure their differences.

The paper separates labor-augmenting productivity, which changes the effective services supplied by labor, from Hicks-neutral productivity, which shifts output for a given bundle of effective inputs. Using a restricted translog production function, I identify these components from input choices and productivity transition equations. I then extend the framework to imperfect competition. The ratio of the labor and materials first-order conditions removes the common markup, providing information about labor-augmenting productivity. Additional dynamic restrictions allow productivity and markup dynamics to be studied jointly. The framework makes explicit the assumptions needed to distinguish technological improvements from market power.

I use Spanish manufacturing data from the Encuesta Sobre Estrategias Empresariales over 2000–2019, constructing a thirteen-variable digitalization index covering information technologies, digital human capital, automation, and stakeholder-facing technologies. Firm-specific output and materials price indices allow expenditures to be deflated. The pooled benchmark estimation uses 17,247 firm-year observations from 2,237 firms, with coverage varying across specifications.

In the perfect-competition benchmark, lagged digitalization is positively associated with both productivity components, with stronger labor-augmenting gains among small firms. Digital human capital is particularly important for labor augmentation. Under imperfect competition, labor-augmenting results remain broadly positive. The average markup response is small, but markups increase in industries with limited digital adoption and decrease in more digitally intensive industries. These findings show why evaluating digitalization requires distinguishing how firms produce from how they price.

Productivity and Industry Dynamics in Banking

In “**The Scale–Technology Paradox: Technological Progress and Productivity Decline in Nigerian Banking**” (with Subal C. Kumbhakar; published in *Economic Modelling*, 2026, article 107859), I examine why technological modernization and consolidation need not translate into higher productivity. Banks produce loans and investments using common resources, so a single-output production function can miss important features of their technology. The paper estimates a multi-output input distance function with a control-function approach grounded in cost minimization. This framework accommodates endogenous input choices and separates productivity change from technical change without requiring input price data.

Using annual data for seventeen Nigerian commercial banks over 2000–2023, we estimate an average annual productivity decline of 3.5 percent, with larger and internationally licensed banks experiencing steeper declines. At the same time, technical change is cost-reducing and estimated returns to scale average 1.46. Lending is the principal source of cost pressure. These findings document a gap between improvements in production possibilities and realized productivity performance. They suggest that technological upgrading and consolidation alone may be insufficient: lending efficiency, asset quality, and the ability to realize scale economies also matter. This project broadens my production-function research to a regulated, multi-output industry and connects productivity measurement to the organization of financial intermediation.

Causal Mechanisms and Flexible Econometric Methods

My working paper “**Mediation Analysis in Production Functions with Unobserved Productivity**” (with Subal C. Kumbhakar; under review at *Economics Letters*) studies the channels through which a treatment such as digitalization affects output. Technology adoption may change output directly and indirectly through firms’ labor choices. Conventional mediation methods can misattribute these channels when unobserved productivity affects adoption, labor demand, and output. Correcting the production equation for endogenous inputs alone does not resolve the confounding in the mediator equation.

The paper develops a sequential estimator that combines an Akerberg–Caves–Frazer-style production-function estimator with a mediator regression controlling for recovered productivity. It derives direct, labor-mediated, and total effects, including specifications in which treatment interacts with labor. A firm-block bootstrap repeats the estimation stages to account for the generated productivity control. Monte Carlo experiments show that the estimator recovers the effects under the model’s assumptions, whereas omitting productivity from the mediator equation can substantially overstate the indirect effect. An application to Spanish manufacturing uses labor as the mediator and lagged digitalization as the treatment. The contribution is a framework for studying economic mechanisms when firms choose the mediator in response to information unavailable to the researcher.

My published paper, “**Causal Forests versus Penalized Splines for Heterogeneous Treatment Effects**” (with Ivan Korolev and Nancy Dhameja, *Economics Letters*, 2026), addresses a complementary question: which flexible methods best recover variation in treatment effects? We compare causal forests with varying-coefficient penalized splines in simulations with randomized treatment assignment. This design isolates the recovery of heterogeneity from the separate problem of adjusting for confounding.

Forests perform best in designs with sharp, localized treatment-effect transitions. Penalized splines generally perform better when heterogeneity is smooth, when the baseline outcome and treatment effect have different structures, and in designs combining continuous, binary, and group-level covariates. As the number of groups rises, regularized group effects help splines remain accurate while forest performance deteriorates. These findings give applied researchers practical guidance about estimator choice and demonstrate the value of matching flexibility to the structure of the empirical problem. They also motivate my ongoing work on methods that preserve economically meaningful quantities rather than optimizing predictive accuracy alone.

Ongoing Work and Future Research

My next projects build on these contributions in two directions. In “**Artificial Intelligence and Spatial Factor-Biased Productivity**” (with Subal C. Kumbhakar), I am extending the digitalization framework to study how AI adoption changes the relative productivity of inputs and how those changes spread across firms and regions. Knowledge flows, worker mobility, and input-output linkages may transmit gains beyond an adopting firm. I plan to compare measures of exposure based on geographic proximity and economic relationships, and use simulations to assess how well the model distinguishes spillovers from common shocks. This work connects within-firm technological change to the evolution of productivity differences across industries and locations.

In my solo project, “**Regularization in Translog Production Models**,” I evaluate Lasso, Ridge, Elastic Net, principal component regression, and partial least squares as responses to multicollinearity in flexible production models. The central criterion is the accuracy and interpretability of output elasticities, which underpin measures of scale and market power. This project asks when regularization can stabilize economically relevant estimates without distorting the production relationships they are intended to recover.

Over the longer term, I plan to connect heterogeneous treatment-effect methods with production models in which adoption and input choices are jointly endogenous. I also intend to develop diagnostics and sensitivity exercises for the timing, persistence, and exclusion restrictions used to recover latent productivity and markups. My goal is to build an empirical research program that explains how technology reshapes firms and industries while making the assumptions behind those explanations transparent and testable.

References

Publications

- [1] “Causal Forests versus Penalized Splines for Heterogeneous Treatment Effects,” with Ivan Korolev and Nancy Dhameja. *Economics Letters*, 268, 113205, 2026. doi:10.1016/j.econlet.2026.113205.
- [2] “The Scale–Technology Paradox: Technological Progress and Productivity Decline in Nigerian Banking,” with Subal C. Kumbhakar. *Economic Modelling*, 107859, 2026. doi:10.1016/j.econmod.2026.107859. Replication code.

Job Market Paper

- [3] “Digitalization, Factor-Biased Productivity, and Market Power,” with Subal C. Kumbhakar. Job market paper, September 2026.

Working Papers

- [4] “Mediation Analysis in Production Functions with Unobserved Productivity,” with Subal C. Kumbhakar. Under review at *Economics Letters*.

Work in Progress

- [5] “Regularization in Translog Production Models.” Solo-authored.
- [6] “Artificial Intelligence and Spatial Factor-Biased Productivity,” with Subal C. Kumbhakar.